Lecture Notes

Lecture 7

Inner Product Spaces

So far our vector spaces have had no geometry: no length, no angle, no notion of two vectors being perpendicular. The *inner product* supplies all three at once, and with it linear algebra becomes the language of quantum mechanics. The overlap $\langle \phi \mid \psi \rangle$ is a probability amplitude; the normalization $\langle \psi \mid \psi \rangle = 1$ fixes total probability; orthogonality $\langle \phi \mid \psi \rangle = 0$ means two pure states can be perfectly distinguished by a suitable projective measurement. This lecture sets up inner products in the *physics convention* and in Dirac's bra–ket notation, and proves the inequality that underlies every amplitude bound in the subject: Cauchy–Schwarz.

A word on convention. We take the inner product to be *conjugate-linear in the first slot and linear in the second*—the standard physics convention, so that $\langle \phi \mid \psi \rangle$ is linear in the ket $|\psi\rangle$. Many mathematics texts put the conjugation in the second slot; every formula converts by swapping the two arguments.

Remark 7.1 (Standing scope for Lectures 7–10). The mathematical core is finite-dimensional over $\mathbb{R}$ or $\mathbb{C}$; quantum examples use complex spaces. A normalized ket is a representative of a pure-state ray, so $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ represent the same physical state. The measurement examples use the finite ideal-projective model, not the most general quantum measurement.

Learning objectives

By the end of this lecture you will be able to:

- Verify the inner-product axioms in the physics convention (conjugate-linear first slot).
- Move fluently between $\langle \cdot, \cdot \rangle$ and Dirac $\langle \cdot \mid \cdot \rangle$ notation.
- Use norms, orthogonality, and the Pythagorean theorem.
- Prove and apply Cauchy–Schwarz and the triangle inequality; use polarization.

1 Inner products

Definition 7.2 (Inner product, physics convention). An *inner product* on a complex vector space V assigns to each ordered pair $u, v \in V$ a scalar $\langle u, v \rangle \in \mathbb{C}$ such that

1. *conjugate symmetry*: $\langle u, v \rangle = \overline{\langle v, u \rangle}$;
2. *linearity in the second slot*: $\langle u, av + bw \rangle = a\langle u, v \rangle + b\langle u, w \rangle$;
3. *positive definiteness*: $\langle v, v \rangle \geq 0$, with equality iff $v = 0$.

A vector space with an inner product is an *inner product space*.

Conjugate symmetry forces $\langle v, v \rangle = \overline{\langle v, v \rangle}$ to be real, so condition (3) makes sense. Combining (1) and (2) gives *conjugate-linearity in the first slot*:

$$\langle au + bw, v \rangle = \overline{\langle v, au + bw \rangle} = \overline{a\langle v, u \rangle + b\langle v, w \rangle} = \overline{a}\,\langle u, v \rangle + \overline{b}\,\langle w, v \rangle.$$

The scalars come out of the first slot conjugated and out of the second slot unchanged—"one and a half" times linear, or *sesquilinear*. Over $\mathbb{R}$ the conjugations vanish and an inner product is just a symmetric, positive-definite bilinear form.

Example 7.3 (The standard inner products). On $\mathbb{C}^n$, the standard inner product is

$$\langle u, v\rangle = \sum_{k=1}^{n} \overline{u_k}\, v_k = u^\dagger v,$$

conjugating the *first* vector's components (here $u^\dagger = \overline{u}^\mathsf{T}$ is the conjugate transpose). On $\mathbb{R}^n$ this is the ordinary dot product $\sum_k u_k v_k$. On the space $C[a,b]$ of continuous functions, where $a < b$,

$$\langle f, g\rangle = \int_a^b \overline{f(t)}\, g(t)\, dt$$

is an inner product. Continuity is load-bearing: if $f(t_0) \neq 0$, then $|f|$ stays bounded away from zero on a small interval, so $\int_a^b |f|^2 > 0$. Thus a zero squared norm forces $f = 0$. This dense but incomplete model is distinct from the later physical $L^2[a,b]$ space, whose vectors are almost-everywhere equivalence classes and which is complete. ◁

Example 7.4 (A computation in $\mathbb{C}^2$). With $u = (1, i)$ and $v = (2,\ 1 - i)$ in $\mathbb{C}^2$,

$$\langle u, v\rangle = \overline{1}\cdot 2 + \overline{i}\cdot(1 - i) = 2 + (-i)(1 - i) = 2 + (-i - 1) = 1 - i,$$

while $\langle v, u\rangle = \overline{2}\cdot 1 + \overline{(1 - i)}\cdot i = 2 + (1 + i)i = 2 + (i - 1) = 1 + i = \overline{\langle u, v\rangle}$, as conjugate symmetry demands. ◁

Example 7.5 (The Frobenius inner product). On $M_n(\mathbb{C})$, $\langle A, B\rangle = \operatorname{tr}(A^\dagger B) = \sum_{j,k} \overline{A_{jk}}\, B_{jk}$ is an inner product; it is the standard inner product on $\mathbb{C}^{n^2}$ in disguise, and it will measure the "size" of operators later. ◁

Example 7.6 (Weighted products and a non-example). For fixed weights $w_1, \ldots, w_n > 0$, $\langle u, v\rangle = \sum_k w_k \overline{u_k} v_k$ is again an inner product on $\mathbb{C}^n$. Positivity is exactly what requires $w_k > 0$: with $w = (1, -1)$ the nonzero vector $(0, 1)$ would have $\langle v, v\rangle = -1 < 0$, so the form fails to be an inner product. The positivity axiom is what makes $\|v\| = \sqrt{\langle v, v\rangle}$ a genuine length. ◁

Proposition 7.7 (Basic facts). In any inner product space, $\langle v, 0\rangle = \langle 0, v\rangle = 0$, and if $\langle u, x\rangle = \langle w, x\rangle$ for every x, then $u = w$.

Proof. $\langle v, 0\rangle = \langle v, 0\cdot 0\rangle = 0\cdot\langle v, 0\rangle = 0$, and $\langle 0, v\rangle = \overline{\langle v, 0\rangle} = 0$. For the second claim, conjugate-linearity in the first slot gives $\langle u - w, x\rangle = 0$ for all x; taking $x = u - w$ yields $\|u - w\|^2 = 0$, so $u = w$. ■

Example 7.8 (Checking the axioms). For the standard product on $\mathbb{C}^n$: conjugate symmetry is $\langle u, v\rangle = \sum_k \overline{u_k} v_k = \overline{\sum_k \overline{v_k} u_k} = \overline{\langle v, u\rangle}$; linearity in the second slot is the distributive law; and $\langle v, v\rangle = \sum_k |v_k|^2 \geq 0$ vanishes only when every $v_k = 0$. For $C[a,b]$ with $a < b$, conjugate symmetry and linearity pass through the integral; the continuity argument in Example 7.3 supplies positive definiteness. ◁

Remark 7.9 (Why the conjugation is not optional). Over $\mathbb{C}$, conjugate symmetry forces $\langle v, v\rangle$ to be real, so the requirement $\langle v, v\rangle \geq 0$ even makes sense; without the conjugation in the first slot, $\langle v, v\rangle$ could be complex and "length" would be undefined. The physics convention keeps amplitudes linear in the ket while guaranteeing real, positive norms.

2 Dirac notation

Physicists write vectors as *kets* $|\psi\rangle$ and pair them using *bras*. To each vector ϕ Dirac associates the bra $\langle\phi|$, the linear functional that sends $|\psi\rangle$ to the inner product:

$$\langle\phi| \colon |\psi\rangle \longmapsto \langle\phi \mid \psi\rangle = \langle\phi, \psi\rangle.$$

Because the inner product is conjugate-linear in its first slot, the correspondence $|\phi\rangle \mapsto \langle\phi|$ is conjugate-linear: $a|\phi\rangle + b|\chi\rangle \mapsto \overline{a}\,\langle\phi| + \overline{b}\,\langle\chi|$. The bracket $\langle\phi \mid \psi\rangle$ is then linear in the ket and conjugate-linear in the bra—exactly Definition 7.2.

The reversed product $|\psi\rangle\langle\phi|$ is an *operator*: it sends $|\chi\rangle$ to $|\psi\rangle\langle\phi \mid \chi\rangle$, a scalar multiple of $|\psi\rangle$. Such rank-one "outer products" build projections and, in Lecture 8, resolutions of the identity.

Remark 7.10 (Physics bridge: the Born rule). For normalized states, $\langle\phi \mid \psi\rangle$ is the *probability amplitude* to find a system prepared in $|\psi\rangle$ to be in $|\phi\rangle$, and $|\langle\phi \mid \psi\rangle|^2$ is the probability. Cauchy–Schwarz below will show this probability never exceeds 1.

Example 7.11 (Physics bridge: an amplitude). Let $|\psi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)$ and $|\phi\rangle = |0\rangle$ in an orthonormal two-state basis $|0\rangle, |1\rangle$ (so $\langle 0 \mid 0\rangle = \langle 1 \mid 1\rangle = 1$, $\langle 0 \mid 1\rangle = 0$). Then $\langle\phi \mid \psi\rangle = \frac{1}{\sqrt{2}}\big(\langle 0 \mid 0\rangle + \langle 0 \mid 1\rangle\big) = \frac{1}{\sqrt{2}}$, so the probability of finding the system in $|0\rangle$ is $|\langle\phi \mid \psi\rangle|^2 = \frac{1}{2}$. ◃

Example 7.12 (Bridge: a rank-one projector as a matrix). With $|0\rangle = (1, 0)$ and $|1\rangle = (0, 1)$, the outer product

$$|0\rangle\langle 0| = \begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}1 & 0\end{pmatrix} = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}$$

is the projection onto $|0\rangle$, and $|0\rangle\langle 0| + |1\rangle\langle 1| = I$ is a *resolution of the identity*. Projections and observables will be assembled from such outer products in the next lectures. ◃

Remark 7.13 (Bridge: finite identity resolutions and expectation values). In a finite-dimensional space with orthonormal basis $\{|1\rangle, \dots, |n\rangle\}$, the finite *resolution of the identity* $\sum_{k=1}^{n} |k\rangle\langle k| = I$ expands any state as $|\psi\rangle = \sum_{k=1}^{n} |k\rangle\langle k \mid \psi\rangle$, with coefficients the amplitudes $\langle n \mid \psi\rangle$. The *expectation value* of an observable A in the state $|\psi\rangle$ is the bra–ket sandwich $\langle A\rangle = \langle\psi|\,A\,|\psi\rangle$, which Lecture 9 will show is real for observables. We construct such bases systematically in Lecture 8.

Example 7.14 (Physics bridge: an expectation value). For $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and the observable $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, since $\sigma_z|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$,

$$\langle\sigma_z\rangle = \langle\psi|\,\sigma_z\,|\psi\rangle = \tfrac{1}{2}\big(\langle 0 \mid 0\rangle - \langle 0 \mid 1\rangle + \langle 1 \mid 0\rangle - \langle 1 \mid 1\rangle\big) = \tfrac{1}{2}(1 - 1) = 0,$$

the average of the outcomes $+1$ and -1, each with probability $\frac{1}{2}$. The bra–ket sandwich computes averages directly. ◃

3 Norm, orthogonality, and the Pythagorean theorem

> **Definition 7.15 (Norm and orthogonality).** The *norm* of v is $\|v\| = \sqrt{\langle v, v\rangle}$. A vector with $\|v\| = 1$ is a *unit vector*; in the quantum examples it is a normalized representative of a pure-state ray, not a distinct physical state for every global phase. Two vectors are *orthogonal*, $u \perp v$, if $\langle u, v\rangle = 0$.

The norm is homogeneous, $\|av\| = \sqrt{\langle av, av\rangle} = \sqrt{\overline{a}\,a\,\langle v, v\rangle} = |a|\,\|v\|$, and vanishes only at 0. The single most useful identity relating norm and orthogonality is the following.

Theorem 7.16 (Pythagorean theorem). If $u \perp v$, then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$.

Proof. Expanding with sesquilinearity,

$$\|u+v\|^2 = \langle u+v, u+v\rangle = \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2.$$

Orthogonality kills the cross terms $\langle u, v\rangle = \langle v, u\rangle = 0$, leaving $\|u\|^2 + \|v\|^2$. ∎

Example 7.17 (Normalizing a state). The vector $|\psi\rangle = (1, i, 1) \in \mathbb{C}^3$ has $\|\psi\|^2 = \langle \psi, \psi\rangle = |1|^2 + |i|^2 + |1|^2 = 3$, so the normalized state is $\frac{1}{\sqrt{3}}(1, i, 1)$, a unit vector describing the same physics. The induced distance $\|u - v\|$ makes V a metric space—the setting for convergence and completeness in Lectures 13–14. ◁

Definition 7.18 (Orthogonal complement). The *orthogonal complement* of a subspace $U \subseteq V$ is $U^\perp = \{v \in V : \langle u, v\rangle = 0 \text{ for all } u \in U\}$, itself a subspace of V.

Proposition 7.19 (Pythagoras for many vectors). If $v_1, \dots, v_m$ are pairwise orthogonal, then $\|v_1 + \cdots + v_m\|^2 = \|v_1\|^2 + \cdots + \|v_m\|^2$.

Proof. Expand $\langle \sum_j v_j, \sum_k v_k\rangle = \sum_{j,k}\langle v_j, v_k\rangle$; every off-diagonal term vanishes by orthogonality, leaving $\sum_j \|v_j\|^2$. ∎

Example 7.20 (Pythagoras with three components). The standard basis vectors of $\mathbb{R}^3$ are pairwise orthogonal, so for $v = (2, -1, 2) = 2e_1 - e_2 + 2e_3$, Proposition 7.19 gives $\|v\|^2 = 2^2 + (-1)^2 + 2^2 = 9$ and $\|v\| = 3$—the Euclidean length read as a sum over orthogonal components, the same bookkeeping that makes $\|\psi\|^2 = \sum_n |\langle n \mid \psi\rangle|^2$ a total probability. ◁

Example 7.21 (Independence via the Gram matrix). The *Gram matrix* of $v_1, \dots, v_m$ has entries $G_{jk} = \langle v_j, v_k\rangle$. The vectors are linearly independent if and only if G is invertible: a dependence $\sum_k c_k v_k = 0$ yields $Gc = 0$ after pairing with each v_j. Conversely, if $Gc = 0$, set $z = \sum_k c_k v_k$; then $\|z\|^2 = c^\dagger G c = 0$, so $z = 0$, and a nonzero kernel vector c is a dependence. For $v_1 = (1, 0)$, $v_2 = (1, 1)$ in $\mathbb{R}^2$, $G = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ has $\det G = 1 \neq 0$, confirming independence. ◁

Example 7.22 (An orthonormal pair). In $\mathbb{C}^2$ the vectors $e_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $e_2 = \frac{1}{\sqrt{2}}(1, -1)$ have $\langle e_1, e_1\rangle = \langle e_2, e_2\rangle = 1$ and $\langle e_1, e_2\rangle = \frac{1}{2}(1 - 1) = 0$: they are orthonormal. Any $u = (a, b)$ then expands as $u = \langle e_1, u\rangle\, e_1 + \langle e_2, u\rangle\, e_2$, the prototype of the orthonormal expansions of Lecture 8. ◁

Example 7.23 (Norms and overlaps of functions). On $C[0, 1]$ with $\langle f, g\rangle = \int_0^1 \overline{f}\, g\, dt$, the function $f(t) = t$ has $\|f\|^2 = \int_0^1 t^2\, dt = \frac{1}{3}$, so $\|f\| = 1/\sqrt{3}$. The functions 1 and t are not orthogonal, since $\langle 1, t\rangle = \int_0^1 t\, dt = \frac{1}{2} \neq 0$; Gram–Schmidt (Lecture 8) will replace them by an orthogonal pair, the first *shifted* Legendre polynomials on $[0, 1]$ (the standard Legendre family is orthogonal on $[-1, 1]$). ◁

4 The Cauchy–Schwarz inequality

Every vector splits into a part along a given direction and a part orthogonal to it; that decomposition proves the central inequality of the subject.

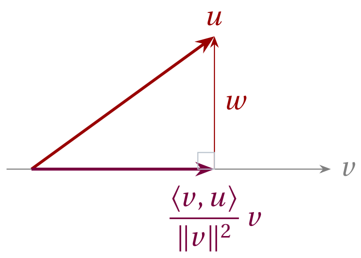

Figure 1: Orthogonal decomposition $u = \frac{\langle v,u\rangle}{\|v\|^2} v + w$ with $w \perp v$. The Pythagorean theorem applied to this right triangle gives Cauchy–Schwarz.

Theorem 7.24 (Cauchy–Schwarz). For all u, v in an inner product space,

$$|\langle u, v\rangle| \le \|u\|\,\|v\|,$$

with equality if and only if u, v are linearly dependent.

Proof. If $v = 0$ both sides vanish. Otherwise write

$$u = \frac{\langle v, u\rangle}{\|v\|^2}\, v + w, \qquad w = u - \frac{\langle v, u\rangle}{\|v\|^2}\, v.$$

A direct check gives $\langle v, w\rangle = \langle v, u\rangle - \frac{\langle v,u\rangle}{\|v\|^2}\langle v, v\rangle = 0$, so $w \perp v$ (Figure 1). The two pieces are orthogonal, so by the Pythagorean theorem

$$\|u\|^2 = \frac{|\langle v, u\rangle|^2}{\|v\|^2} + \|w\|^2 \ge \frac{|\langle v, u\rangle|^2}{\|v\|^2}.$$

Multiplying by $\|v\|^2$ and taking square roots gives $|\langle v, u\rangle| \le \|u\|\|v\|$; since $|\langle u, v\rangle| = |\langle v, u\rangle|$, the inequality follows. Equality holds iff $\|w\| = 0$, i.e. $w = 0$, i.e. u is a scalar multiple of v. ∎

Remark 7.25 (The same proof, algebraically). Cauchy–Schwarz is often presented as positivity of a quadratic: for $v \neq 0$ set $\lambda = \langle v, u\rangle/\|v\|^2$ and expand

$$0 \le \|u - \lambda v\|^2 = \|u\|^2 - \frac{|\langle v, u\rangle|^2}{\|v\|^2},$$

which rearranges to the inequality at once. This is the orthogonal decomposition of Figure 1 seen through algebra rather than geometry; the vector $u - \lambda v$ is exactly the orthogonal part w.

Example 7.26 (Cauchy–Schwarz, checked). For $u = (1, i)$, $v = (2, 1 - i)$ of Example 7.4, $|\langle u, v\rangle| = |1 - i| = \sqrt{2}$, while $\|u\| = \sqrt{1+1} = \sqrt{2}$ and $\|v\| = \sqrt{4+2} = \sqrt{6}$. Indeed $\sqrt{2} \le \sqrt{2}\cdot\sqrt{6} = 2\sqrt{3}$, with strict inequality because u, v are independent. ◁

Example 7.27 (Orthogonalizing a pair). The decomposition behind the proof gives a recipe: for independent v_1, v_2, the vector $w_2 = v_2 - \frac{\langle v_1,v_2\rangle}{\|v_1\|^2} v_1$ is orthogonal to v_1. For $v_1 = (1, 0)$, $v_2 = (1, 1)$ in $\mathbb{R}^2$ this yields $w_2 = (1, 1) - 1\cdot(1, 0) = (0, 1)$. Iterating the step over a whole basis is precisely the Gram–Schmidt process of Lecture 8. ◁

Remark 7.28 (Physics bridge: amplitudes are bounded). For unit vectors the Cauchy–Schwarz inequality reads $|\langle \phi \mid \psi\rangle| \le 1$: no transition amplitude between normalized states exceeds 1 in magnitude, so the Born probability $|\langle \phi \mid \psi\rangle|^2$ is genuinely a probability. The same inequality, applied to operators acting on a state, yields the Heisenberg uncertainty relation (Lecture 9).

Remark 7.29 (Classical inequalities for free). Applied to the function and Frobenius inner products of Examples 7.3 and 7.5, Cauchy–Schwarz immediately yields

$$\left|\int_a^b \overline{f} g \, dt\right|^2 \le \int_a^b |f|^2 \, dt \int_a^b |g|^2 \, dt, \qquad |\mathrm{tr}(A^\dagger B)|^2 \le \mathrm{tr}(A^\dagger A)\ \mathrm{tr}(B^\dagger B).$$

These integral and matrix estimates are awkward to prove head-on but fall out of the one abstract inequality—a payoff of thinking about inner products structurally rather than componentwise.

5 Triangle inequality, parallelogram law, polarization

Theorem 7.30 (Triangle inequality). For all u, v, $\|u + v\| \le \|u\| + \|v\|$.

Proof. Using $\langle u, v\rangle + \langle v, u\rangle = 2\,\mathrm{Re}\langle u, v\rangle$ and then Cauchy–Schwarz,

$$\|u + v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u, v\rangle + \|v\|^2 \le \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2 \le \big(\|u\| + \|v\|\big)^2.$$

Taking square roots completes the proof. ∎

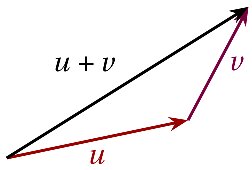

Figure 2: The triangle inequality: the direct path $u + v$ is no longer than the two-leg path of lengths $\|u\|$ and $\|v\|$.

Two further identities express the rigidity of inner-product geometry. Expanding $\|u \pm v\|^2$ and adding or subtracting gives the parallelogram law and the polarization identity.

Proposition 7.31 (Parallelogram law and polarization). For all u, v in a complex inner product space,

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2,$$

and the inner product is recovered from the norm by

$$\langle u, v\rangle = \tfrac{1}{4} \sum_{\alpha \in \{1, -1, i, -i\}} \alpha\, \|v + \alpha u\|^2.$$

Proof. Expanding $\|u \pm v\|^2 = \langle u \pm v, u \pm v\rangle$ by sesquilinearity gives $\|u \pm v\|^2 = \|u\|^2 + \|v\|^2 \pm 2\Re\langle u, v\rangle$; adding the two sign choices cancels the cross term and gives the parallelogram law. For polarization, the same expansion with $|\alpha| = 1$ gives, for each $\alpha \in \{1, -1, i, -i\}$,

$$\|v + \alpha u\|^2 = \|v\|^2 + \|u\|^2 + \alpha\langle v, u\rangle + \overline{\alpha}\,\langle u, v\rangle.$$

Multiplying by α and summing, the relations $\sum_\alpha \alpha = 0$, $\sum_\alpha \alpha^2 = 0$, and $\sum_\alpha \alpha\overline{\alpha} = 4$ leave $\sum_\alpha \alpha\, \|v + \alpha u\|^2 = 4\langle u, v\rangle$. ∎

The parallelogram law characterizes which norms come from an inner product. On $\mathbb{R}^n$ or $\mathbb{C}^n$ with $n \ge 2$, among the standard p-norms for $1 \le p \le \infty$, exactly the 2-norm is induced by an inner product. In one dimension every standard p-norm equals absolute value and comes from the standard inner product.

Remark 7.32 (Optional extension: the Jordan–von Neumann converse). The nontrivial converse—that a norm satisfying the parallelogram law has a polarization formula that defines an inner product—is the Jordan–von Neumann theorem. It is a useful characterization, but is not part of the assessed core here.

The polarization identity shows that the inner product carries no information beyond lengths. In the real case it becomes $\langle u, v\rangle = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2)$, and the angle between *nonzero* vectors is defined by

$$\cos\theta = \frac{\langle u, v\rangle}{\|u\|\,\|v\|},$$

which Cauchy–Schwarz guarantees lies in $[-1, 1]$. In a complex space the overlap generally has a phase and there is no corresponding real-angle formula; its normalized magnitude instead controls transition probability.

Example 7.33 (An angle and the parallelogram law). In $\mathbb{R}^2$, $u = (1, 0)$ and $v = (1, 1)$ have $\langle u, v\rangle = 1$, $\|u\| = 1$, $\|v\| = \sqrt{2}$, so $\cos\theta = \frac{1}{\sqrt{2}}$ and $\theta = 45°$. The parallelogram law checks: $\|u+v\|^2 + \|u-v\|^2 = \|(2,1)\|^2 + \|(0,-1)\|^2 = 5 + 1 = 6 = 2\|u\|^2 + 2\|v\|^2$. ◁

Example 7.34 (Recovering the product from lengths). For the same real u, v, polarization gives $\langle u, v\rangle = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2) = \frac{1}{4}(5-1) = 1$, matching the direct value: the inner product is encoded entirely in the lengths. ◁

Example 7.35 (Distance and the triangle inequality). For $u = (1, i)$ and $v = (2, 1-i)$ of Example 7.4, the distance is $\|u - v\| = \|(-1,\ 2i-1)\| = \sqrt{1+5} = \sqrt{6}$, and the triangle inequality $\sqrt{6} \le \|u\| + \|v\| = \sqrt{2} + \sqrt{6}$ holds comfortably. Even in complex space distances obey the familiar geometry. ◁

Example 7.36 (A norm with no inner product). The max norm $\|(x,y)\|_\infty = \max(|x|, |y|)$ on $\mathbb{R}^2$ violates the parallelogram law: for $u = (1,0)$, $v = (0,1)$, $\|u+v\|_\infty^2 + \|u-v\|_\infty^2 = 1 + 1 = 2$, but $2\|u\|_\infty^2 + 2\|v\|_\infty^2 = 4$. By Proposition 7.31 no inner product can induce $\|\cdot\|_\infty$: inner-product geometry is genuinely special among norms. ◁

6 Physics bridge: states, amplitudes, and what comes next

In the finite pure-state model, a unit vector $|\psi\rangle$ represents a ray, since an overall phase is unobservable. The amplitude $\langle\phi\,|\,\psi\rangle$ governs interference, and $|\langle\phi\,|\,\psi\rangle|^2$ is the Born probability for the rank-one projective outcome $|\phi\rangle$. Orthogonal pure states ($\langle\phi\,|\,\psi\rangle = 0$) can be distinguished with certainty by a suitable projective measurement; an arbitrary measurement need not distinguish them. For a finite-dimensional self-adjoint observable, distinct scalar outcomes are eigenvalues and their spectral eigenspaces are orthogonal (Lecture 9); the outcomes themselves are numbers, not vectors.

Example 7.37 (Extension: interference of qubit states). Let $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. They are orthonormal, since $\langle + \,|\, -\rangle = \frac{1}{2}(\langle 0\,|\,0\rangle - \langle 1\,|\,1\rangle) = 0$, so a system in $|+\rangle$ is never found in $|-\rangle$. Yet each has overlap $\frac{1}{\sqrt{2}}$ with $|0\rangle$, so a measurement in the $\{|0\rangle, |1\rangle\}$ basis returns either outcome with probability $\frac{1}{2}$. The relative phase distinguishing $|+\rangle$ from $|-\rangle$ is invisible to that measurement but decisive in others—the algebra of interference. ◁

Remark 7.38 (Optional extension: Cauchy–Schwarz and uncertainty). Cauchy–Schwarz is the engine of the Heisenberg uncertainty relation. For observables A, B and a state $|\psi\rangle$, set $\Delta A = A - \langle A\rangle I$ and $\Delta B = B - \langle B\rangle I$. Cauchy–Schwarz applied to the vectors $\Delta A|\psi\rangle$ and $\Delta B|\psi\rangle$ gives

$$\langle(\Delta A)^2\rangle\,\langle(\Delta B)^2\rangle \ge |\langle\psi|\,\Delta A\,\Delta B\,|\psi\rangle|^2 \ge \left(\tfrac{1}{2}|\langle[A,B]\rangle|\right)^2.$$

With $[\hat{x}, \hat{p}] = i\hbar I$ this becomes $\Delta x\, \Delta p \geq \hbar/2$: a bound on amplitudes turns into a bound on simultaneous knowledge. The full argument waits for adjoints in Lecture 9.

Remark 7.39 (Bridge: from finite spaces toward Hilbert space). A complex inner product space that is also *complete*—every Cauchy sequence of vectors converges to a vector in the space—is a *Hilbert space*. Every finite-dimensional inner product space is automatically complete, so the distinction is invisible here; the infinite-dimensional state spaces of quantum mechanics, such as L^2, require completeness as a genuine extra axiom, taken up in Lecture 13.

Example 7.40 (Bridge: closest-point projection preview). Among all scalar multiples of a unit vector e, the one closest to a given u is $\langle e, u\rangle\, e$: the error $u - \langle e, u\rangle\, e$ is orthogonal to e, so by Pythagoras $\|u - ce\|^2 = \|u - \langle e, u\rangle\, e\|^2 + |c - \langle e, u\rangle|^2$ is smallest at $c = \langle e, u\rangle$. This orthogonal projection is the principle behind least-squares fitting and the orthonormal expansions of Lecture 8. ◁

That raises the finite-dimensional question driving the next lecture: can we choose a basis of mutually orthogonal unit vectors—an *orthonormal basis*—and expand any vector as a finite sum in it? The answer is yes, by the Gram–Schmidt procedure, and in such a basis the inner product becomes the familiar sum $\sum_k \overline{u_k} v_k$ and operators acquire especially clean matrices. Hilbert bases, norm-convergent series, and infinite identity resolutions wait until Lecture 13.

Summary of main concepts

An inner product is sesquilinear (conjugate-linear in the first slot, linear in the second, in the physics convention), conjugate-symmetric, and positive definite; in Dirac notation $\langle \phi \mid \psi \rangle$ is linear in the ket and conjugate-linear in the bra. It induces a norm $\|v\| = \sqrt{\langle v, v\rangle}$, orthogonality $\langle u, v\rangle = 0$, and the Pythagorean theorem. The Cauchy–Schwarz inequality $|\langle u, v\rangle| \leq \|u\|\|v\|$ follows from orthogonal decomposition and bounds every amplitude; from it come the triangle inequality and, with the parallelogram law and polarization identity, the full geometry of the space. In the finite pure-state model, normalized vectors represent rays, $|\langle \phi \mid \psi \rangle|^2$ is a transition probability, and a suitable projective measurement distinguishes orthogonal pure states—the setting in which the finite spectral theorem will live.

- **Inner product (physics convention)**—sesquilinear, conjugate-symmetric, positive-definite; $\langle \phi \mid \psi \rangle$ is linear in the ket.
- **Norm and orthogonality**—$\|v\| = \sqrt{\langle v, v\rangle}$; $\langle u, v\rangle = 0$; the Pythagorean theorem.
- **Cauchy–Schwarz**—$|\langle u, v\rangle| \leq \|u\|\,\|v\|$; bounds every amplitude.
- **Triangle inequality, parallelogram law, polarization**—the metric geometry of the space; recover $\langle \cdot, \cdot \rangle$ from $\|\cdot\|$.
- **Finite pure-state picture**—normalized vectors represent rays; $|\langle \phi \mid \psi \rangle|^2$ is a transition probability; a suitable projective measurement distinguishes orthogonal states.

Exercises for this lecture are in the companion sheet `exercises-7.pdf`.